

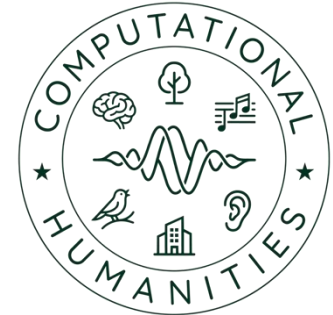
University of Bamberg



Master Project - Machine Listening (CH-Proj-M)

01 – Literature Research

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Lecture Structure



Week	Day	Project Phase	Topics
1	13.04.		Overview, master project organization, open project topics
2	20.04.	1. Literature research	Literature research strategies, reference management
3	27.04.		Basics: academic writing, required structure of the term paper
4	04.05. (P1)	2. Data acquisition (Dataset research, data recording)	Role of data and annotations in machine listening research, data protection, licences (CC etc.)
5	11.5.		Basics: audio editing and annotation (Audacity, Sonic Visualizer)

Literature Research

Motivation



- What is already known about the topic?
 - Research always builds on existing knowledge
 - Avoid repeating what has already been done
- Identify gaps, disagreements, or unanswered questions in prior studies
 - Derive a clear direction & purpose of your research
- This guides the choice of methods, theories, and concepts
 - What has been used before? (strengths vs. weaknesses)

Literature Research

Recipe



- Clearly define the topic and scope of your literature review to focus on relevant literature
- Use multiple databases and sources to ensure comprehensive coverage

Literature Research

Recipe



- Think about keywords and synonyms for search
- Example Topic:
 - “Few-Shot Learning for Rare Bioacoustic Event Detection”
- Possible search key words
 - “few-shot learning”
 - “few-shot learning audio”
 - “sound event detection”
 - “bioacoustics”

Literature Research

Recipe



- Think about keywords and synonyms for search
- Example Topic:
 - “Bird Activity Detection in Audio”
- Possible search key words
 - “bird activity detection”
 - “audio segmentation”
 - “bioacoustic”

Literature Research

Recipe



- Critically evaluate the sources for their relevance, credibility, and quality
 - Check [impact factor of conferences](#)

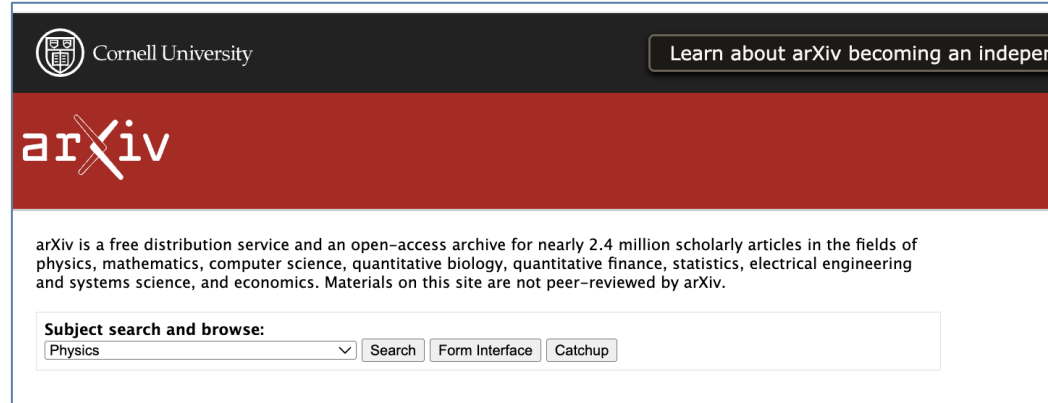
	Publication	h5-index	h5-median
1.	IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)	137	203
2.	Conference of the International Speech Communication Association (INTERSPEECH)	112	160
3.	IEEE/ACM Transactions on Audio, Speech, and Language Processing	79	131
4.	Journal of Sound and Vibration	74	91
5.	Applied Acoustics	70	96
6.	IEEE Transactions on Ultrasonics, Ferroelectrics, and Frequency Control	54	75
7.	The Journal of the Acoustical Society of America	53	66

Literature Research

Recipe



- Preprints (arXiv)
 - Unpublished work (papers, articles, reports)
 - Often currently in peer-review process



Literature Research

Recipe



- Preprints (arXiv)
 - Check whether pre-print has been “properly” published at a conference / journal in the meantime
 - If yes, chose this reference instead

Literature Research Recipe



arXiv > cs > arXiv:2511.07156

Computer Science > Machine Learning

[Submitted on 10 Nov 2025]

Conditional Diffusion as Latent Constraints for Controllable Symbolic Music Generation

Matteo Pettenò, Alessandro Ilic Mezza, Alberto Bernardini

Recent advances in latent diffusion models have demonstrated state-of-the-art performance in high-dimensional time-series data synthesis while providing flexible guidance. However, existing methodologies primarily rely on musical context or natural language as the main modality of interacting with the generative process, where users seek precise fader-like control over specific musical attributes. In this work, we explore the application of denoising diffusion processes as plug-and-play late symbolic music generation models. We focus on a framework that leverages a library of small conditional diffusion models operating as implicit probabilistic priors backbone. While previous studies have explored domain-specific use cases, this work, to the best of our knowledge, is the first to demonstrate the versatility of such musical attributes, such as note density, pitch range, contour, and rhythm complexity. Our experiments show that diffusion-driven constraints outperform traditional latent constraints architectures, achieving significantly stronger correlations between target and generated attributes while maintaining high perceptual quality and

Subjects: **Machine Learning (cs.LG)**; Artificial Intelligence (cs.AI); Audio and Speech Processing (eess.AS)

Cite as: arXiv:2511.07156 [cs.LG]

(or arXiv:2511.07156v1 [cs.LG] for this version)

<https://doi.org/10.48550/arXiv.2511.07156>

Journal reference: In Proc. of the 26th International Society for Music Information Retrieval Conference (ISMIR 2025), Daejeon, Korea, Sept. 21–25, 2025

Submission history

From: Matteo Pettenò [\[view email\]](#)

[v1] Mon, 10 Nov 2025 14:46:10 UTC (7,285 KB)

Cite ISMIR paper instead (peer-reviewed)!

Literature Research Sources



- <https://scholar.google.de/>

The screenshot shows the Google Scholar search results for the query "acoustic anomaly detection survey". The search bar at the top shows the query and a magnifying glass icon. Below the search bar, it indicates "Ungefähr 55.000 Ergebnisse (0,14 Sek.)". On the left side, there are filters for "Artikel" and a sidebar with options like "Beliebige Zeit", "Nach Relevanz sortieren", "Nach Datum sortieren", "Beliebige Sprache", "Alle Typen", "Patente einschließen", "Zitate einschließen", and "Alert erstellen". The main results list includes three entries:

- [HTML] Anomalous sound event detection: A survey of machine learning based methods and applications** [HTML] springer.com
Z Mnasri, S Rovetta, F Masulli - Multimedia Tools and Applications, 2022 - Springer
... This **survey** is focused on **anomaly detection**, so it covers specifically those SED applications where the challenge is to recognize **anomalous** events rather than to segment an **acoustic** ...
☆ Speichern 99 Zitieren Zitiert von: 42 Ähnliche Artikel Alle 7 Versionen
- Anomalous sound detection with machine learning: A systematic review** [PDF] arxiv.org
EC Nunes - arXiv preprint arXiv:2102.07820, 2021 - arxiv.org
... **anomaly detection**, semi-supervised **anomaly detection**, and ... Thus, this **survey** can enable a general analysis of the scope ... Real time **detection of acoustic anomalies** in industrial ...
☆ Speichern 99 Zitieren Zitiert von: 26 Ähnliche Artikel Alle 2 Versionen 99
- Anomaly detection: A survey** [PDF] acm.org
V Chandola, A Banerjee, V Kumar - ACM computing surveys (CSUR), 2009 - dl.acm.org
... Most of the existing **surveys** on **anomaly detection** either focus on a particular application ... **surveys** discuss **anomaly detection** techniques that **detect** the simplest form of **anomalies**. ...
☆ Speichern 99 Zitieren Zitiert von: 14370 Ähnliche Artikel Alle 34 Versionen

Below these, there is another entry:

- Acoustic anomaly detection using convolutional autoencoders in industrial processes** [PDF] researchgate.net
TB Duman, B Bayram, G Ince - ... Conference on Soft Computing Models in ..., 2020 - Springer
... In the **survey** paper [7], various **anomaly detection** works and fault ... In this study, a CAE based end-to-end **acoustic anomaly detection** method was proposed which can be used within the ...
☆ Speichern 99 Zitieren Zitiert von: 46 Ähnliche Artikel Alle 3 Versionen

Literature Research Sources



- <https://ieeexplore.ieee.org>

Showing 1-25 of 345 results for **acoustic anomaly detection** ×

☐ Conferences (262) ☐ Journals (74) ☐ Magazines (8) ☐ Early Access Articles

Search

Show

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☐ Subscribed Content ?
☐ Open Access Only

Year

☒ Range ☐ Single Year

1980 2024

Author ▾
Affiliation ▾
Publication Title ▾

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Get notified when new research is published matching your search criteria.

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☐ **Design and development of an acoustic sensor array for anomaly detection** 

Dibyendu Roy; V. Ramu Reddy; Parijat Deshpande; Ranjan Dasgupta
2016 10th International Conference on Sensing Technology (ICST)
Year: 2016 | Conference Paper | Publisher: IEEE
Cited by: [Papers \(1\)](#)

Abstract [HTML](#)  

☐ **Marine mammal sound anomaly and quality detection using multitaper spectrogram and hydrophone big data** 

Farook Sattari; Colter McQuay; Peter F. Driessen
2017 IEEE Pacific Rim Conference on Communications, Computers and Signal Processing (PACRIM)
Year: 2017 | Conference Paper | Publisher: IEEE

Abstract [HTML](#)  

Literature Research Sources



- <https://arxiv.org/>

Showing 1–50 of 50 results for all: acoustic anomaly detection

Search v0.5.6 released 2020-02-24 [Feedback?](#)

acoustic anomaly detection All fields Search

☒ Show abstracts ☐ Hide abstracts

[Advanced Search](#)

50 results per page. Sort results by Announcement date (newest first) Go

1 2

[Next](#)

1. [arXiv:2403.00379](#) [[pdf](#), [other](#)] [eess AS](#) [cs SD](#)

The Impact of Frequency Bands on **Acoustic Anomaly Detection** of Machines using Deep Learning Based Model

Authors: Tin Nguyen, Lam Pham, Phat Lam, Dat Ngo, Hieu Tang, Alexander Schindler

Abstract: In this paper, we propose a deep learning based model for **Acoustic**... [More](#)

Submitted 1 March, 2024; originally announced March 2024.

2. [arXiv:2402.07691](#) [[pdf](#), [other](#)] [cs RO](#)

Evaluation of a Smart Mobile Robotic System for Industrial Plant Inspection and Supervision

Authors: Georg K. J. Fischer, Max Bergau, D. Adriana Gómez-Rosal, Andreas Wachaja, Johannes Gräter, Matthias Odenweller, Uwe Piechottka, Fabian Hoeflinger, Nikhil Gosala, Niklas Wetzels, Daniel Büscher, Abhinav Valada, Wolfram Burgard

Abstract: ...system designed for comprehensive plant inspection. This innovative system comprises a robotic platform equipped with a diverse array of sensors integrated to facilitate the **detection** of various process and infrastructure parameters. These sensors encompass optical (LiDAR, Stereo, UV/IR/RGB cameras), olfactory (electronic nose), and... [More](#)

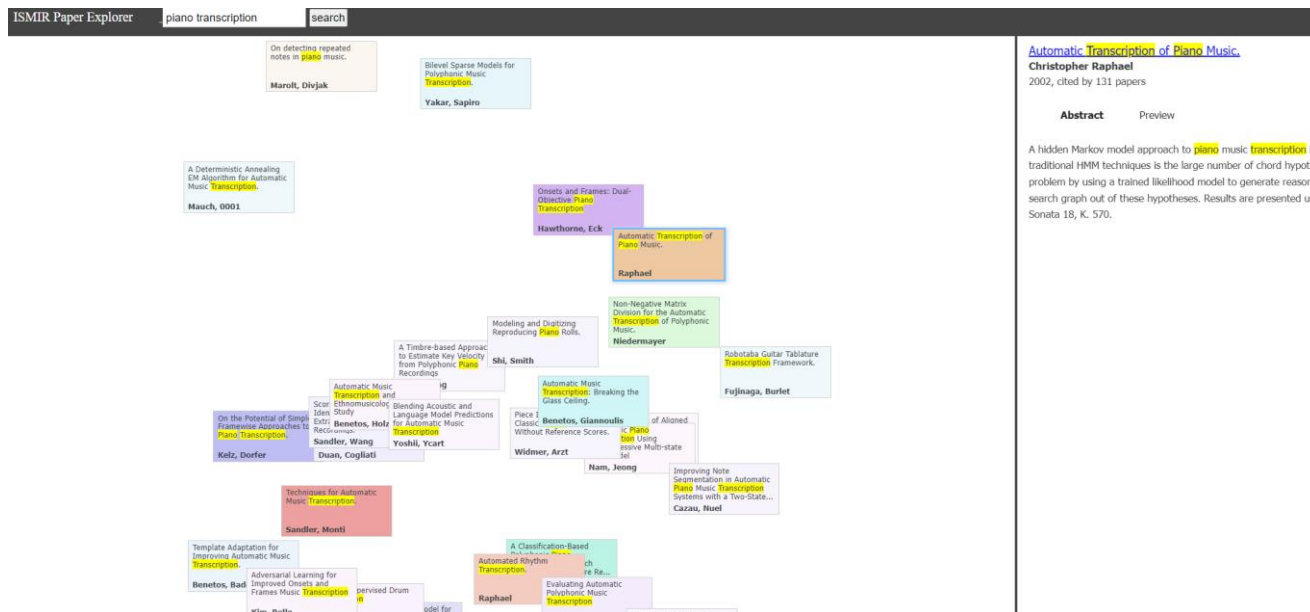
Submitted 12 February, 2024; originally announced February 2024.

Comments: Submitted for publication in IEEE Sensors Journal

Literature Research Sources



- <https://ismir-explorer.ai.ovgu.de/app/>



Literature Research

General Recommendations



- Focus first on most recent publications (2020-2025)
- Reverse literature research (check references of references)

Sound Event Detection: A Tutorial

Annamaria Mesaros¹, Toni Heittola¹, Tuomas Virtanen¹, Mark D. Plumbley²

¹ Tampere University, Finland

email: {annamaria.mesaros, toni.heittola, tuomas.virtanen}@tuni.fi

² University of Surrey, UK

email: m.plumbley@surrey.ac.uk



- [9] T. Heittola, A. Mesaros, T. Virtanen, and A. Eronen, "Sound event detection in multisource environments using source separation," in *Workshop on Machine Listening in Multisource Environments, CHiME2011*, Florence, Italy, 2011, pp. 36–40.
- [10] A. Mesaros, T. Heittola, and D. Ellis, "Datasets and evaluation," in *Computational Analysis of Sound Scenes and Events*, T. Virtanen, M. D. Plumbley, and D. Ellis, Eds. Springer, 2018, ch. 6, pp. 147–179.
- [11] E. Fonseca, M. Plakal, D. P. Ellis, F. Font, X. Favory, and X. Serra, "Learning sound event classifiers from web audio with noisy labels," in *2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. IEEE, 2019, pp. 21–25.

Survey article, 2021

Literature Research

General Recommendations



- Articles are around 8-12 pages -> more comprehensive review
 - Look for “review paper” / “survey paper”

SPRINGER NATURE Link

Find a journal Publish with us Track your research Search

Home > [Multimedia Tools and Applications](#) > Article

A parametric survey on polyphonic sound event detection and localization

Published: 02 August 2024

Volume 84, pages 22083–22120, (2025) [Cite this article](#)

A Survey of Evaluation in Music Genre Recognition

Bob L. Sturm

Audio Analysis Lab, AD:MT, Aalborg University Copenhagen
A.C. Meyers Vænge 15, DK-2450 Copenhagen SV, Denmark
bst@create.aau.dk

Literature Research

General Recommendations



- Focus on conference papers (high-ranked conferences:
 - [ISMIR](#)
 - [DAFx](#),
 - [ICMC](#),
 - IEEE [ICASSP](#) / [WASPAA](#)
- and journals
 - [TISMIR](#)
 - IEEE [TASLP](#)

Literature Research

How to (quickly) read a paper?

- First skim through (fast read-through)
 - Introduction
 - Conclusion
 - Figures / Tables
- Second (more detailed) read (if promising)
 - Methodology
 - Evaluation & Results

Literature Research

How to take notes?



- Mark passages using PDF reader (Acrobat Reader)
- Save reference metadata as bibtex file (be thorough once, and relaxed later)
- Rename PDF files consistently (e.g. using the BibTex key)
- Take bullet point notes in text file (or LaTeX project)

Literature Research

How to take notes?

- Think of categories/criteria to compare different methods
- This can give your related work section a structure later (one paragraph for each column)
- Example

	Feature	Dataset	Model
Author1	MFCC	AudioSet	CNN
Author2	Mel Spec	AudioSet	Transformer
Author3	Mel Spec	FSD50k	CRNN

Literature Research

How to write a related work section?

- Prioritize peer-reviewed journal articles and reputable publishers
- Categorize the literature into themes or methodological approaches relevant to your research question
- Synthesize information from references: discuss overall trends, major findings, and gaps in the literature
 - Highlight how these findings relate to your research

Literature Research

How to write a related work section?



- Discuss the strengths and limitations of previous studies and identify areas for further research
- Relate the discussed literature to your research question or hypothesis (how did these studies have influenced, supported, or motivated your research?)
- Where applicable, use tables, figures, or flowcharts to summarize and compare key information from the references

Thank you for your attention!

Any questions?



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<https://www.uni-bamberg.de/ch/>